Deploying Mass Scale Molecular Testing

Randy True, CEO FloodLAMP Biotechnologies
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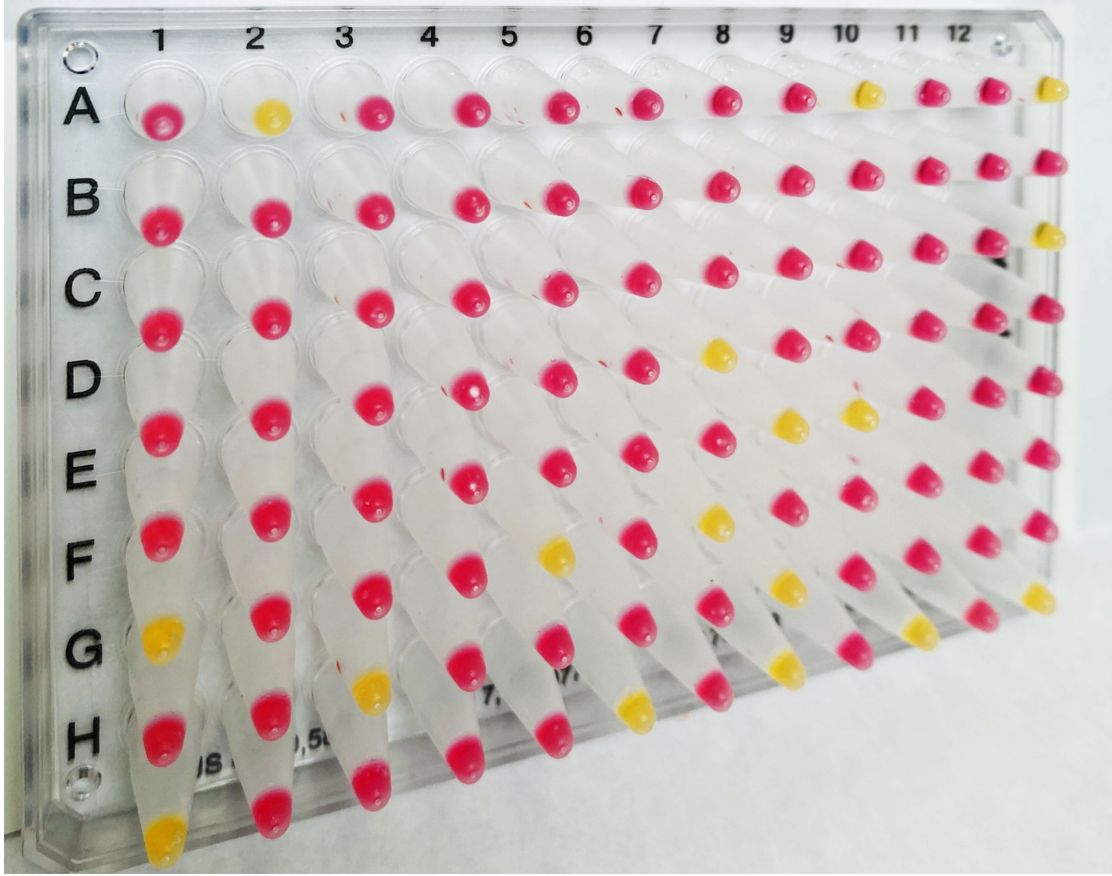
This is what you want to see!

Pink is not detected.

Last yellow is positive control.

Today's Results

Pooled collection of as AN swabs.
Surveillance testing in pop up FloodLAMP lab.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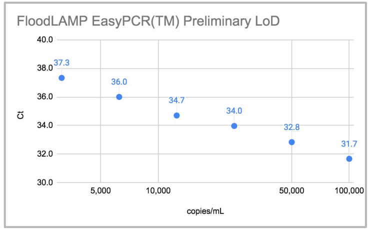
EasyPCR™ Test

- 3 copies/μl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives



FloodLAMP SwabDirect PCR Result	Comparator Positive	Comparator Negative	Total
Positive	39	0	39
Negative	1	40	41
Invalid	0	0	0
Total	40	40	80
Positive Agreement	97.5% (39/40) 95% CI: 86.8% to 99.9%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

QuickColor™ LAMP Test

- 12 copies/μl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

FloodLAMP QuickColor Test Result	Comparator Positive	Comparator Negative	Total
Positive	36	0	36
Negative	4	40	44
Total	40	40	80
Positive Agreement	90.0% (36/40) 95% CI: 76.3% to 97.2%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

Clinical Evaluation Successful

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the "really straightforward" protocol.

Thank You!

Peter Antevy, Eagles and FTFC for invitation and your service.

FloodLAMP
Biotechnologies

Randy True
Founder and CEO
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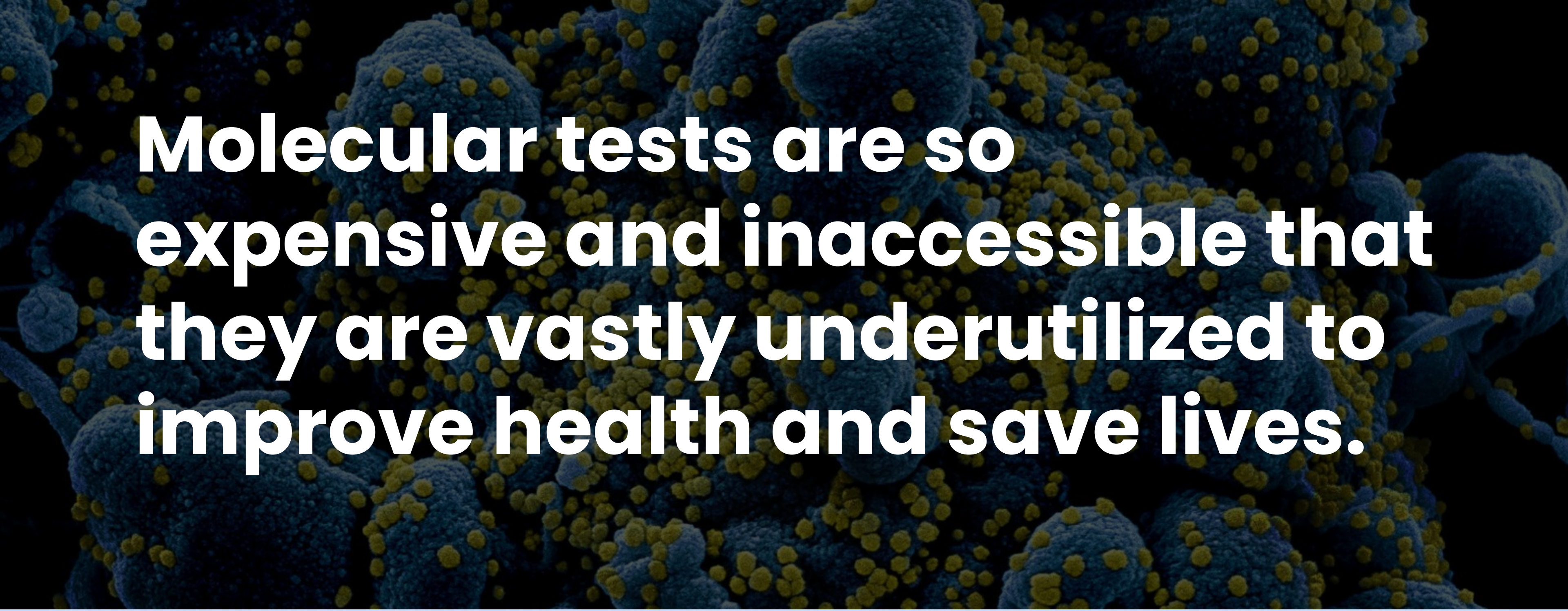
Accessible Testing for the Genomics Age

FloodLAMP
Biotechnologies

FloodLAMP helps people to take control of their health and lets labs grow into new public health markets by unlocking access to molecular testing. Our core product is a mass screening system built around a highly scalable, instrument-free LAMP test, a home collection kit, and a novel digital health app.



Molecular tests are so expensive and inaccessible that they are vastly underutilized to improve health and save lives.

Problem

The current diagnostics model is built around expensive, doctor prescribed, insurance reimbursable tests. This does not scale. The COVID crisis has exposed critical problems in testing, however they also have negative consequences far beyond the pandemic.

During COVID

- Deploy FloodLAMP best-in-class mass screening program
- Focus on COVID-19 school screening processed by independent labs
- Grow lab network, sales channels and FloodLAMP adoption
- Use volume to reduce FloodLAMP adoption friction

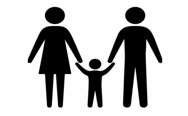
Post COVID

- Key enabler for emerging multi-target molecular screening market
- Apply digital, processing and sourcing infrastructure to high volume clinical and non-clinical molecular applications
- Low prevalence, high consequence screening for public health intelligence, designed for the genomics age
- Excel as an operator built for industry cost compression and commoditization via suite of enabling technologies

Opportunity

Huge govt investment in COVID-19 testing has created a once in a lifetime opportunity for rapid adoption of new modalities. FloodLAMP is using the opportunity for impact and growth in the short term, and as a beachhead to lead a future of accessible molecular testing using commodity chemistries.

How FloodLAMP Screening Works



1) Collect Samples
- pooled swabs
- at-home with family/household
- on-site with pod
- FloodLAMP app to register samples
- fast and easy

2) Return Samples
- pick up or courier to lab/processing site
- low overhead and friction for organizations

3) Same Day Results
- app for students and parents
- web portal for school admins

Solution

FloodLAMP's COVID-19 screening program is an end-to-end solution that makes it easy and affordable to protect interacting groups such as schools, offices, or factories by providing sensitive and rapid turnaround testing.

FloodLAMP in the Lab

Distributed Lab/Sites
- reach to underserved communities
- almost all labs already have needed equipment
- local control
- same day turnaround times
- expandable and replicable

Streamlined Workflow
- simple direct assays
- minimal hands-on time
- modular design
- instrument free
- highly efficient operationally

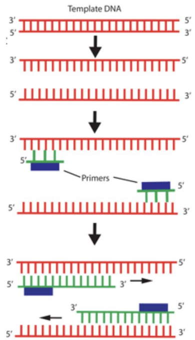
Molecular Tests
- high sensitivity and specificity
- rapid results from LAMP
- deconvolute with LAMP or PCR
- target flexibility

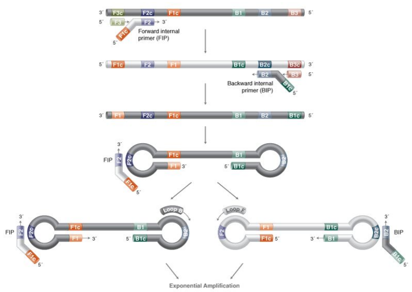
Advantages

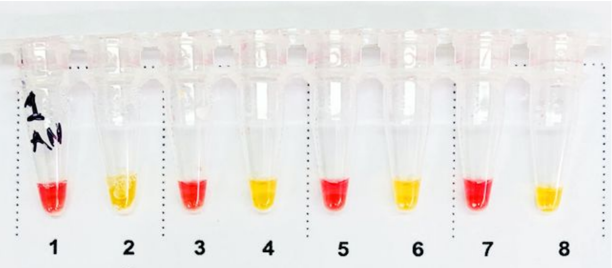
FloodLAMP has licensed best-in-class LAMP amplification chemistry and sample prep from Harvard Medical School. Our streamlined workflow is optimized for mass screening for COVID-19 and beyond.

LAMP: Loop Mediated Isothermal Amplification

- Detects viral RNA
- LAMP generates 10X more DNA in 1/3 time compared to PCR
- Easy visual / camera-based readout

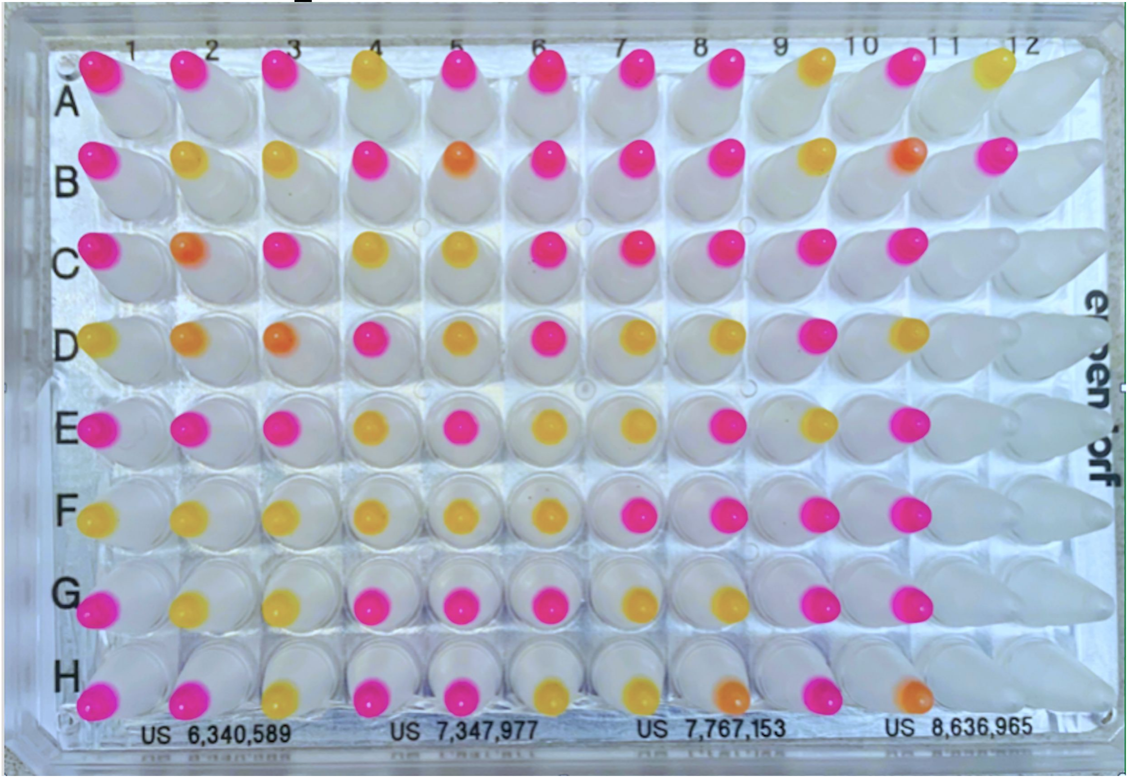


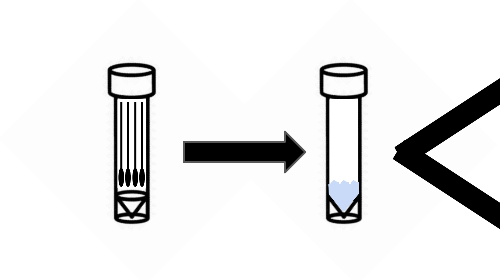




	qPCR	LAMP
Equipment	qPCR machine (thermal cycler & optics)	simple heat block or water bath (isothermal)
Time	90 min	30 min
Capital Cost	$30K-100K	$300

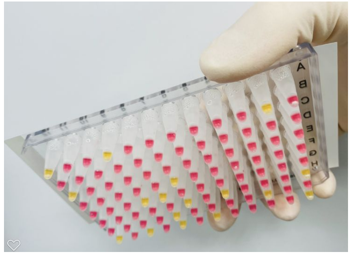
Clinical Sample Plate





Streamlined Sample Prep
- Upfront swab pooling
- Highly scalable, integrated processing
- Same sample for both tests

QuickColor™ LAMP Test
- High sensitivity (90%)
- Ultra-high throughput
- Ideal for serial screening
- Uniquely scales without capital intensive instruments



Visual Readout

EasyPCR™ Test
- Very high sensitivity (98%)
- Medium throughput (1.5 hrs/94)
- Ideal for diagnostics/reflex/confirm

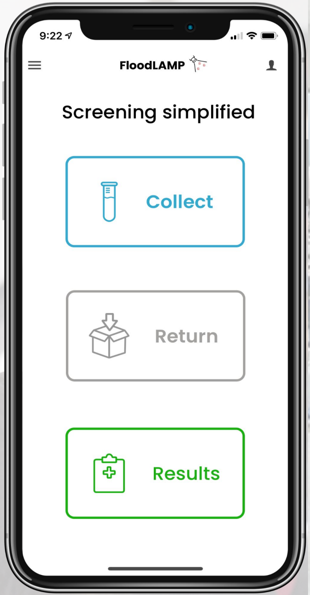
Standard RT-qPCR instrument

Core Assay Technology

FloodLAMP has fully validated 2 complementary tests that are best-of-breed. Emergency Use Authorizations for the tests have been submitted to the FDA, along with the game-changing FloodLAMP Pooled Swab Collection Kit DTC, for interactive review. The IRB for the clinical studies has been approved. The test workflow and collection kit has been designed to expand to non COVID-19 targets in the future.

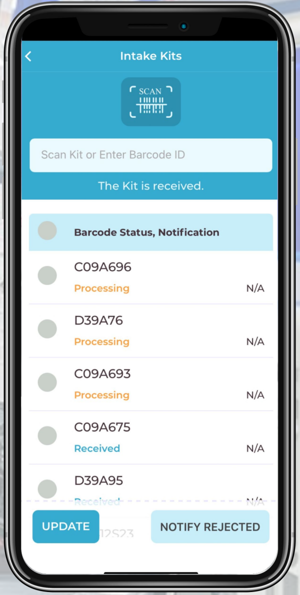
Easy, smooth UI for participants.

- Easy, shareable sign up
- Electronic consent
- Anywhere pooled collection
- Results notifications and reporting



Powerful, lightweight tool for the lab.

- Rapid QR code scanning
- Pre-accessioned, pre-pooled sample tubes
- Batch flow control
- Can replace LIS
- Can interface w LIS thru API



Core Digital Technology

FloodLAMP's digital suite is anchored on a powerful mobile tool for complete screening program management.

Antigen Test Suppliers

Abbott
BD
QUIDEL

Pooled Central PCR

labcorp
BioReference Laboratories an OPKO Health Company
GINKGO BIOWORKS THE ORGANISM COMPANY
color
ThermoFisher SCIENTIFIC
BROAD INSTITUTE
Veritas The Genome Company

	Sensitivity	Time to Results	Prevent Cases in School
FloodLAMP	✓✓✓✓	✓✓✓✓	✓✓✓✓
Antigen	✓✓	✓✓✓✓✓	✓✓
Centralized Lab Processing (pooled PCR)	✓✓✓✓✓	✓✓	✓✓

FloodLAMP provides rapid, high sensitivity results with distributed local labs and processing sites.

Poor sensitivity of antigen tests means they need to be run every day to protect a school, adding large cost and overhead.

The > 24hr turnaround on a classroom PCR pool is a deal breaker for most schools.

"The key is it's all about turnaround time ... [instead of just the lab time] we're redefining turnaround time to be swab to results."
- Mara Aspinall, Advisor and lead author on Rockefeller Foundation National Testing Plan Report

Testing Landscape

FloodLAMP's solution hits the sweet spot for mass screening. Low-sensitivity antigen tests and long turnaround times for centralized lab tests mean more pre-symptomatic, infectious cases in schools. FloodLAMP brings high performance molecular tests to a much broader range of labs.

FloodLAMP Leadership Team

Randy True Founder & CEO

- *Founder TMI Inc. - Acq. Affymetrix 25MM*
- *VP of R&D at Affymetrix*

Kevin Schallert COO

- *Co-Director covid19sci.org, National Volunteer Scientist DB*
- *Founder & COO of VineEye*

Theresa Ling UX Design

- *Design Lead at Uber and New Relic*

Research Aid Networks

Jeremy Rossman ED, Virologist

Volunteers

Vincent Law, Daria Tames, Amanda

Scientific Advisory Board

Anne Wyllie *Yale, Lead Researcher - SalivaDirect*

Prof. Connie Cepko *Harvard, Harvard Medical School, Genetics Department, Co-Dir Trans Med Program, HHMI*

Bill Hyun *UC San Francisco, Genoa Ventures*

Industry Advisors

Tim Lugo *William Blair Biotechnology Group Head*

Zarak Khurshid *Asymmetry Capital*, *Top MDx Analyst*

Michael Hite *Parachute Therapeutics*, *Founder*

John Edge *Oxford Internet Institute, Blue Field Labs, ID2020*

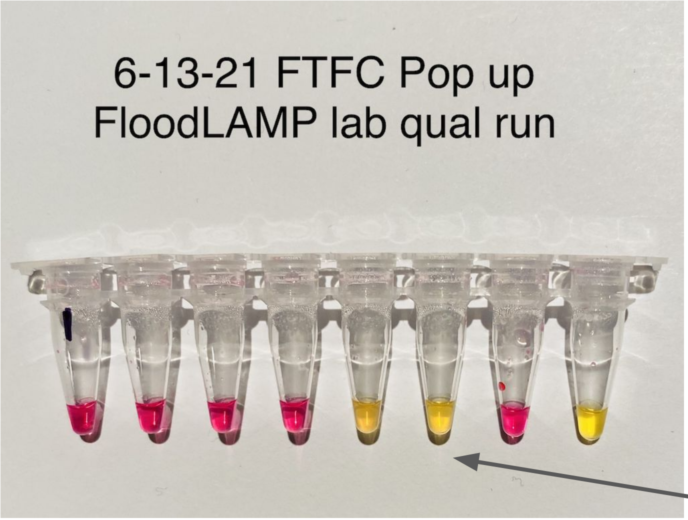
Collaborators

gLAMP Group *Global LAMP consortium of 200+ Academic and industry scientists*

OpenCOVIDScreen Jeff Huber (*Google, Grail*), Cliff Wang (*former Stanford*)

Team

Proven biotech and software experience
Multiple startup exits
Large network of influential advisors and collaborators



Yellow is positive (detected).

Pink is negative (not detected).

Full process controls with spiked inactivated virus.

Qualification Run

Pooled collection of as AN swabs.
Surveillance testing in pop up FloodLAMP lab.
3 hours from showing up with luggage to results.

Molecular Assay Atlas

PEOPLE

Individual

Pool: 5-10

Pool of Pools: 20-100

SAMPLE

Nasopharyngeal Swab (NP)

Cheek Swab

Anterior Nares Swab (AN)

Saliva

INACTIV

TCEP (Chem) + Heat
Rabe Cepko, HUDSON

Proteinase K (Enzyme) + Heat
Saliva Direct

Heat Only

Commercial
Zymo, Lucigen

PURIFIC

Glass Milk
Rabe Cepko

Mag Beads
Kellner, Yu

Skip (Direct)

Commercial
Thermo, Zymo, Kingfisher, Qiagen

AMPLIF

qPCR

LAMP
Colorimetric
Fluorimetric

Misc
RPA, CRISPR, etc.

READOUT

qPCR Machine
Fluorescence

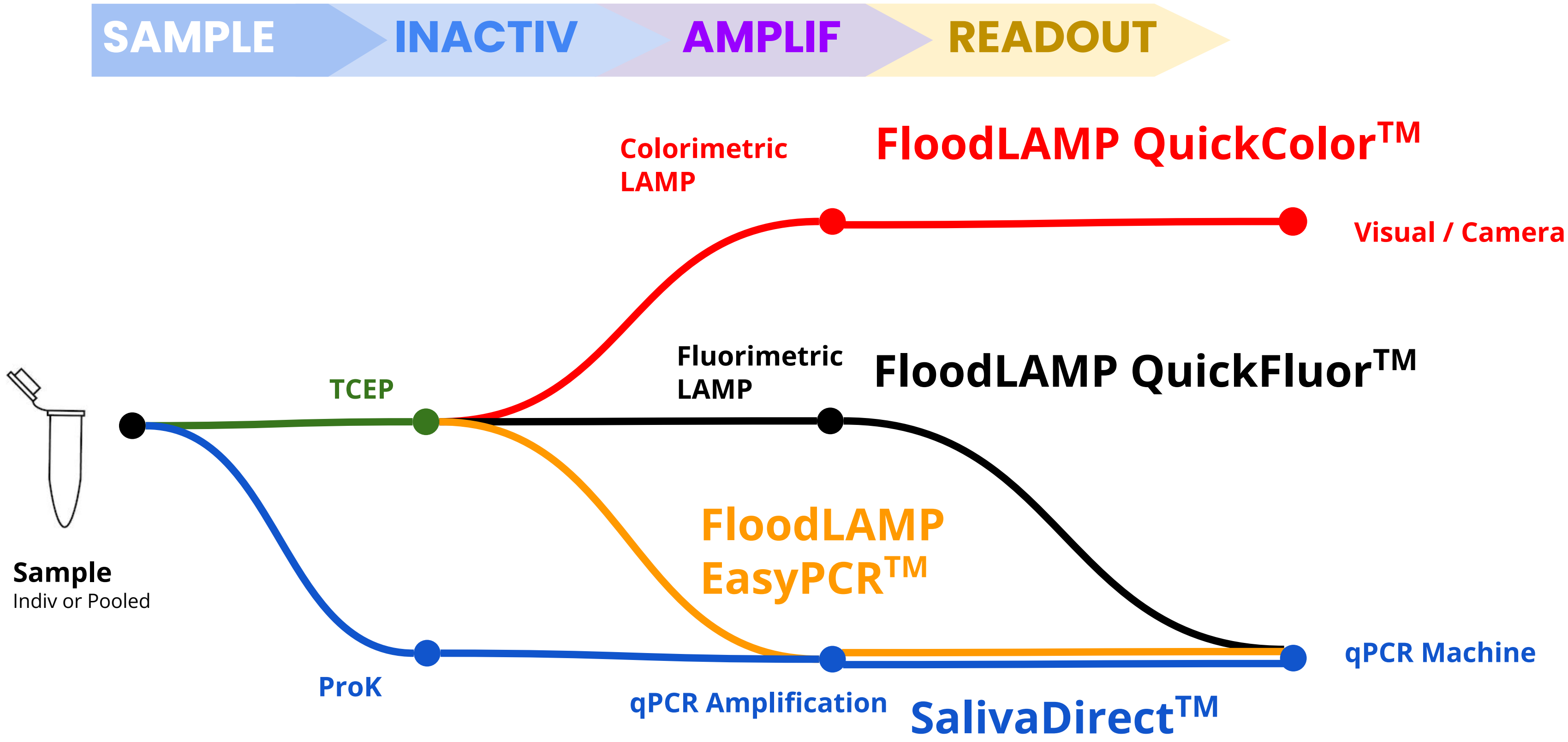
Visual / Camera
Color Change

Plate Reader
Absorbance
Color Genomics

Lateral Flow Strip

FloodLAMP Biotechnologies,
a Public Benefit Corporation
contact@floodlamp.bio

FloodLAMP Tests



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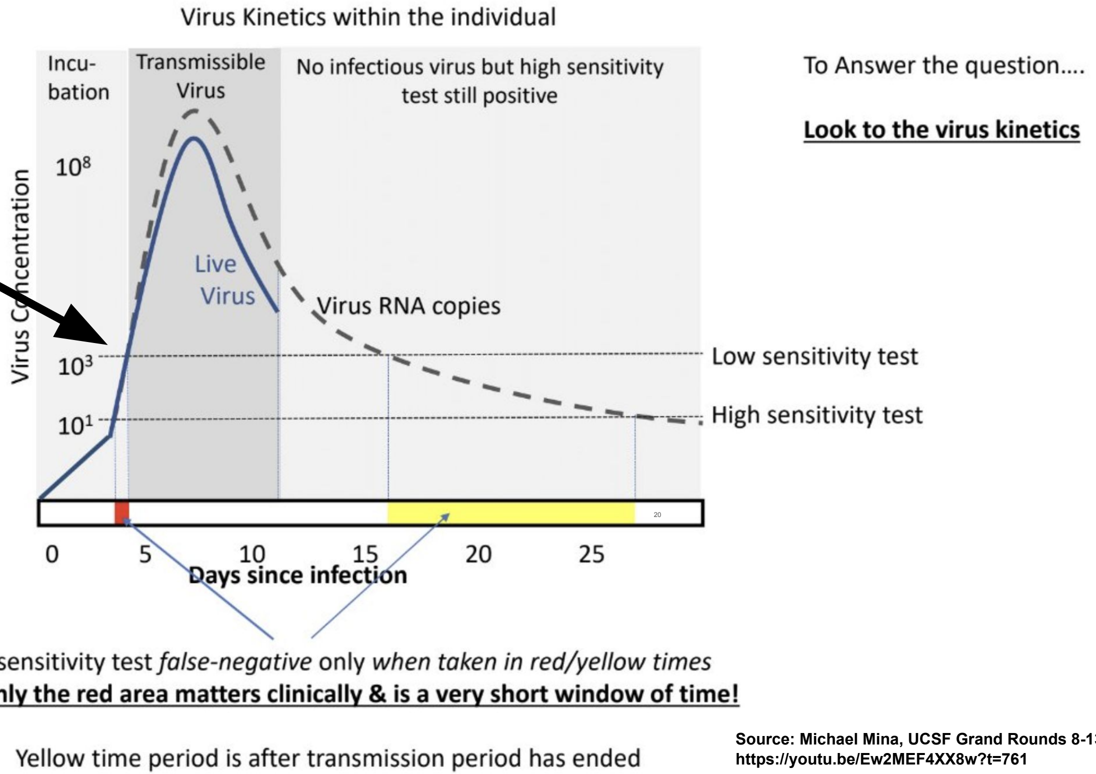
Viral Load and Infectious Detection

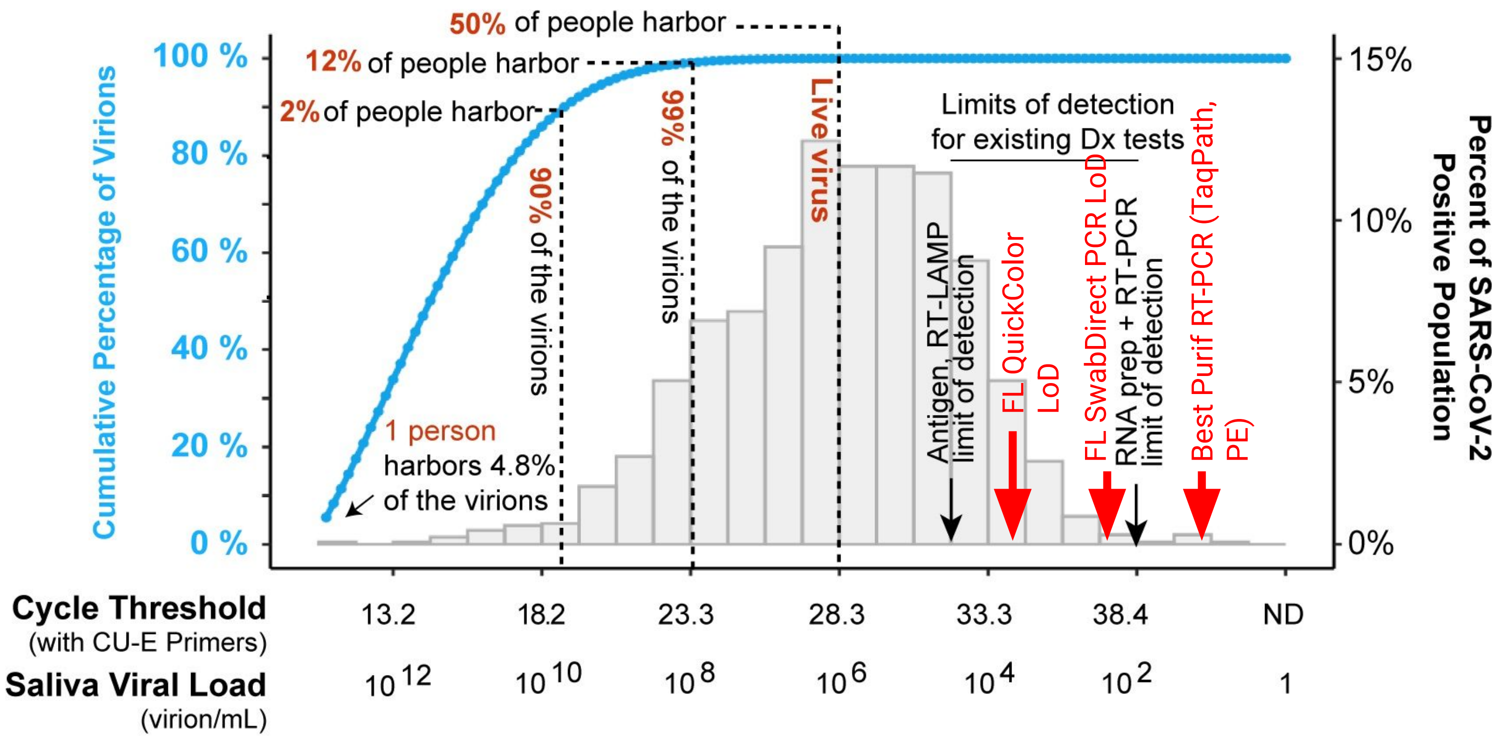
Threshold for "Capable to Infect Others" ~ 1e6/ml

Pooling uses assay sensitivity to scale up coverage - and detect unknown infectious people in interacting populations



Source: Michael Mina, UCSF Grand Rounds 8-13-20
https://youtu.be/Ew2MEF4XX8w?t=761

Viral Load and Infectious Detection



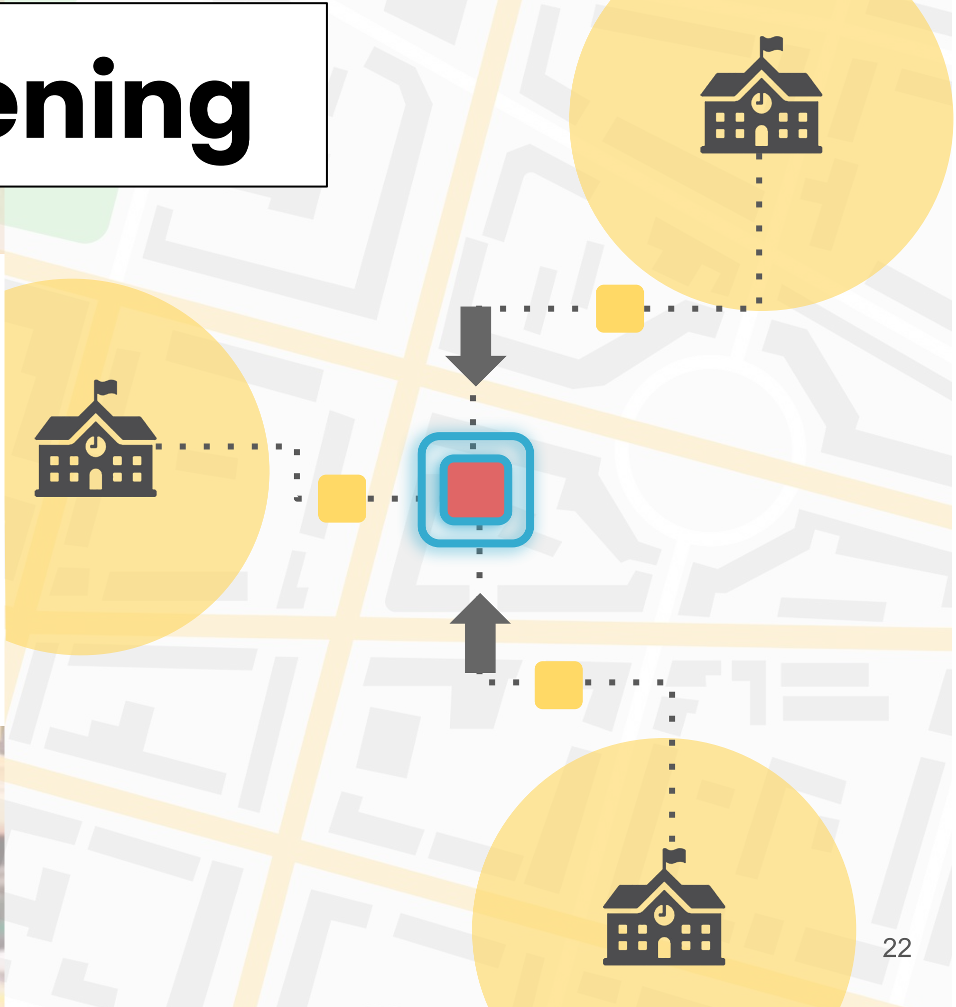
TaqPath Ct	7	12	17	22	27	32
FL Direct PCR	14	19	24	29	34	39

Source: Sara Sawyer Preprint
https://doi.org/10.1101/2021.03.01.21252250

FL School Screening

1. At-Home Pooled Samples
2. Collected at Local Schools
3. Processed by FloodLAMP Network of Labs

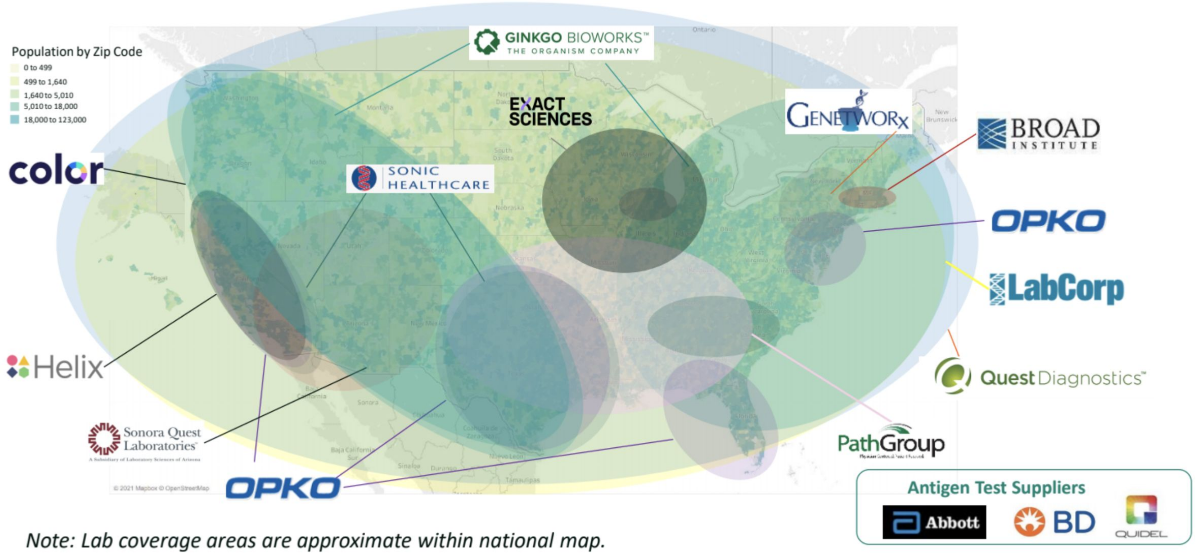
Same Day Results



Current Screening Providers

Includes incumbent and upstart players. Products vary by sensitivity, level of integration and regional focus.

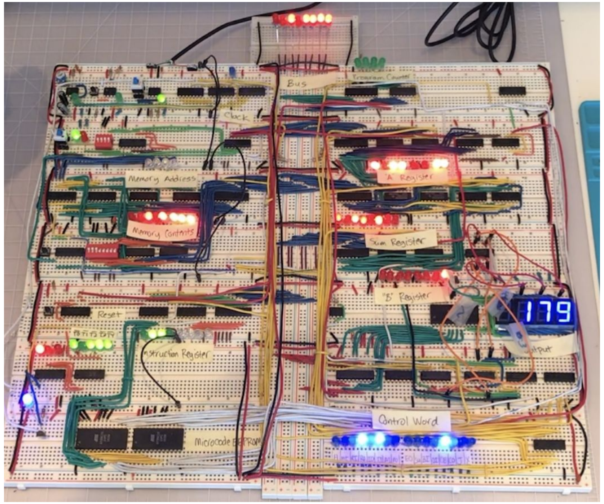
WHO PROVIDES TESTING: LAB FOOTPRINT FOR SCALING K-12 TESTING

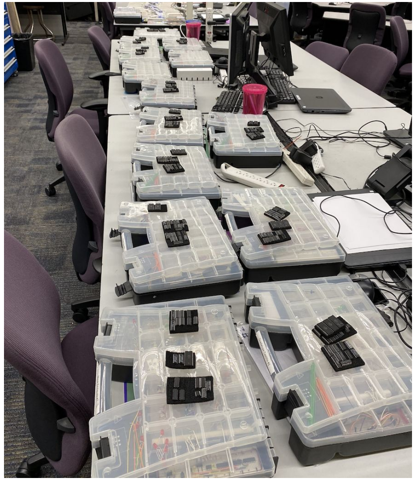


Note: Lab coverage areas are approximate within national map.
Additional coverage possible through public health and university labs

The Rockefeller Foundation · Health Catalysts Group

My CTE Classroom





Regulatory Pathways

1) FDA EUA - Emergency Use Authorization (IVD)
    - LDT for labs and IVD for manufacturers

2) CLIA Lab LDT - Lab Developed Test

3) IRB - Internal Review Board
    - typically at universities
    - needed for EUAs

4) Non-Dx Surveillance
    - not under CLIA
    - no "individual diagnostic" results
    - reflex positives to CLIA test

Regulatory Challenges & Hurdles

- Complexity of regulatory space
- FDA approves test systems not protocols
- Accessing clinical samples
- Performing clinical studies
- Submission paperwork
- Changing priorities

Open EUAs

	Typical IVD EUA	CDC EUA	Yale SalivaDirect™	SHIELD	FloodLAMP EasyPCR™	FloodLAMP QuickColor™
Disclosure of all chemicals and reagents?	No	Yes	Yes	Yes	Yes	Yes
Chemical and reagents available from multiple vendors?	No	Yes	Yes	No	Yes	Yes, *except NEB LAMP MM
Disclosure of primer sequences?	No	Yes std for PCR	Yes	No ProptryThermo	Yes	Yes
Primers commercially available from multiple vendors?	No	Yes	Yes CDC Primers	No ProptryThermo	Yes CDC SD Primers	Yes Available but not launched
Supply chain robust?	No	No/Maybe	Yes	No/Maybe	Yes	Yes
EUA Sponsor Organization Type	For Profit Company	Govt	Academic Not for Profit	Academic Not/For Profit ?	Public Benefit Corp	Public Benefit Corp
Designation of CLIA labs	Kit Sales	N/A open RoR	Impact & Expansion	Impact & Expansion	Impact & Expansion	Impact & Expansion

LAMP EUAS

EUA Date	Test	Diagnostic	Target	IC	Detection	Extraction	Amplification	LOD spike	LOD GE/ 3ml swab
Nov-17	Lucira (At Home POC)	Lucira COVID-19 All-In-One Test Kit	N, N	Lysis Ctrl + IC	Colorimetric	~Direct	Lucira device	Inactivated virus	2,700*
Oct-05	Seasun (2 duplex)	AQ-TOP COVID-19 Rapid Detection Kit PLUS	orf1ab, N	Human RNaseP	PNA Probe	Manual (Qiagen_60704 or Seasun_SS-1300) or Automated (Panagene_PNAK-1001 on PanaMax48)	BioRad_CFX96 or ABI7500	NCCP_43326 genomic RNA	3,000
Sep-01	Detectachem	MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit	N, E	only external controls	Colorimetric	Direct (1µl of transport media into LAMP)	heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or …)	Twist_MT007544.1 synthetic gRNA	225,000
Aug-31	Mammoth	SARS-CoV-2 DETECTR Reagent Kit	N	Human RNaseP	CRISPR Probe	Automated (Qiagen_955134 on Qiagen_EZ1AdvancedBenchtop)	ABI7500	SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA	60,000
Aug-13	Pro-Lab	Pro-AmpRT SARS-CoV-2 Test	RdRP	only external controls	Probe	Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000)	Optigene_GenieHT	BEI_NR-52287 inactivated virus	125*
Jul-09	UCSF/Mammoth	SARS-CoV-2 RNA DETECTR Assay	N	Human RNaseP	CRISPR Probe	automated (Qiagen_955134 on Qiagen_EZ1)	ABI7500	SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA	60,000
May-21	Seasun (1 duplex)	AQ-TOP COVID-19 Rapid Detection Kit	orf1ab	Human RNaseP	PNA Probe	manual (Qiagen_60704)	BioRad_CFX96 or ABI7500	NCCP_43326 gRNA	21,000
May-18/ Aug31	Color	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay	N, E, nsp3	Human RNaseP	Colorimetric	automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360)	plate reader (Biotek_NEO2), Hamilton_Star	ATCC_VR-1986D (gRNA)	2,250
May-06	Sherlock BioSci	Sherlock CRISPR SARS-CoV-2 Kit	orf1ab, N	Human RNaseP	CRISPR Probe	Manual (ThermoFIsher_12280050)	heat block or qPCR, Plate reader (BioTek_NEO2)	gRNA	20,250
Apr-10	Atila BioSystems	iAMP COVID-19 Detection Kit	orf1ab, N	Human GAPDH	Probe	Direct (swab into 350µl, 15min RT then 3µl to LAMP)	BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600	SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus)	3,500*

*Lucira, Pro-Lab and Atila tests LOD/swab uses EUA specified volumes which are less than 3ml

from Matt McFarlane mdmcfar@uvic.ca @mattmcfar

University of Victoria


LAMP!
RAPID RESULTS
New test for koala chlamydia
The University Of Queensland Australia
Create Change
QUT
QUT